CPSC 459 | Project 2 Extra Credit

Consensus agreement under varying network parameters

Experimental method: 54 combinations (3 graph probabilities x 3 malicious-node probabilities x 3 initial transaction distributions x 2 round counts), each evaluated in 50 randomized runs, for 2,700 total runs. A trial is counted successful when every compliant node returns the same nonempty set of transaction IDs.

Strict reporting rule: A malicious-node probability is listed as observed tolerated only if all 50 trials passed for the given remaining parameters. "None" means none of the tested levels (15%, 30%, 45%) met this criterion. This is an empirical agreement metric, not proof of Byzantine fault tolerance or independent validation of transaction correctness.

Maximum observed malicious-node percentage (50/50 criterion)

p_graph	p_txDistribution	numRounds	Maximum observed p_malicious
0.10	0.01	10	None
0.10	0.01	20	None
0.10	0.05	10	None
0.10	0.05	20	None
0.10	0.10	10	None
0.10	0.10	20	None
0.20	0.01	10	None
0.20	0.01	20	None
0.20	0.05	10	None
0.20	0.05	20	15%
0.20	0.10	10	None
0.20	0.10	20	15%
0.30	0.01	10	15%
0.30	0.01	20	15%
0.30	0.05	10	15%
0.30	0.05	20	15%
0.30	0.10	10	15%
0.30	0.10	20	15%

Interpretation

Eight of the 54 tested combinations passed in all 50 trials. For p_graph = 0.10, no tested malicious-node level achieved 50/50 agreement. For p_graph = 0.20, 15% passed at p_txDistribution = 0.05 and 0.10 with 20 rounds. For p_graph = 0.30, 15% passed for all six combinations of transaction distribution and rounds. No 30% or 45% setting passed all 50 trials.

These outcomes show greater observed agreement with denser trust graphs and lower malicious-node probabilities. Randomized runs may vary; a 50/50 outcome is not a guarantee for future networks.

Complete experimental outcomes (successes out of 50)

p_graph	p_malicious	p_txDistribution	Rounds	Successes	Rate
0.10	0.15	0.01	10	20/50	40%
0.10	0.15	0.01	20	17/50	34%
0.10	0.15	0.05	10	21/50	42%
0.10	0.15	0.05	20	21/50	42%
0.10	0.15	0.10	10	23/50	46%
0.10	0.15	0.10	20	23/50	46%
0.10	0.30	0.01	10	0/50	0%
0.10	0.30	0.01	20	0/50	0%
0.10	0.30	0.05	10	0/50	0%
0.10	0.30	0.05	20	0/50	0%
0.10	0.30	0.10	10	0/50	0%
0.10	0.30	0.10	20	0/50	0%
0.10	0.45	0.01	10	0/50	0%
0.10	0.45	0.01	20	0/50	0%
0.10	0.45	0.05	10	0/50	0%
0.10	0.45	0.05	20	0/50	0%
0.10	0.45	0.10	10	0/50	0%
0.10	0.45	0.10	20	0/50	0%
0.20	0.15	0.01	10	46/50	92%
0.20	0.15	0.01	20	49/50	98%
0.20	0.15	0.05	10	47/50	94%
0.20	0.15	0.05	20	50/50	100%
0.20	0.15	0.10	10	47/50	94%
0.20	0.15	0.10	20	50/50	100%
0.20	0.30	0.01	10	12/50	24%
0.20	0.30	0.01	20	9/50	18%
0.20	0.30	0.05	10	8/50	16%
0.20	0.30	0.05	20	10/50	20%
0.20	0.30	0.10	10	11/50	22%
0.20	0.30	0.10	20	14/50	28%
0.20	0.45	0.01	10	0/50	0%
0.20	0.45	0.01	20	0/50	0%

p_graph	p_malicious	p_txDistribution	Rounds	Successes	Rate
0.20	0.45	0.05	10	0/50	0%
0.20	0.45	0.05	20	0/50	0%
0.20	0.45	0.10	10	0/50	0%
0.20	0.45	0.10	20	0/50	0%
0.30	0.15	0.01	10	50/50	100%
0.30	0.15	0.01	20	50/50	100%
0.30	0.15	0.05	10	50/50	100%
0.30	0.15	0.05	20	50/50	100%
0.30	0.15	0.10	10	50/50	100%
0.30	0.15	0.10	20	50/50	100%
0.30	0.30	0.01	10	34/50	68%
0.30	0.30	0.01	20	27/50	54%
0.30	0.30	0.05	10	26/50	52%
0.30	0.30	0.05	20	24/50	48%
0.30	0.30	0.10	10	30/50	60%
0.30	0.30	0.10	20	32/50	64%
0.30	0.45	0.01	10	1/50	2%
0.30	0.45	0.01	20	0/50	0%
0.30	0.45	0.05	10	1/50	2%
0.30	0.45	0.05	20	0/50	0%
0.30	0.45	0.10	10	0/50	0%
0.30	0.45	0.10	20	0/50	0%

Methodological limitation: The checker excludes nodes labeled malicious and compares transaction IDs across compliant nodes, requiring a nonempty result. It does not establish that accepted transactions are correct or that the network resists every adversarial behavior. The observed number of malicious nodes varies by random trial because p_malicious is a probability.